Deciding Presburger Arithmetic Using Automata

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Structure of presentation

- Recall
  - Finite Automata
  - Presburger Arithmetic
- Sets of Integers recognized by a Finite Automaton
- Presburger Formulas to Finite Automata
  - Automata Associated with equalities
  - Automata Associated with inequalities
- Closure Properties and Automata for arbitrary formulas
- Deciding satisfiability of a formula

Finite Automata

- A Finite Automaton is a tuple $(Q,\sum,\delta,q_o,F)$.
  - Q = Finite set of states
  - $\sum$ = Finite set of input symbols
  - $\delta \subseteq (Q \text{x} \sum \text{x} Q)$
  - $q_o \in Q$ (the start state)
  - $F \subseteq Q$ (Finite set of final states)
- A run of the automaton on the word $\omega \in \sum^*$ is a word $\rho \in Q^*$, such that:
  - $\rho[1] = q_o$
  - If $\rho[\text{i}] = p$ and $\rho[\text{i+1}] = q$, then $(p, \omega[\text{i}],q) \in \delta$
- A successful run ρ on ω is when $\rho[\text{n+1}] \in \text{F}$, where $\text{n}=|\omega|$
- A string is accepted by the automaton when there is a successful run on that string.
- Class of languages accepted by finite automaton is closed under union, intersection and complementation.

Presburger Arithmetic

- A Basic term in presburger arithmetic consists of first order variable, constants 0 & 1 and sums of basic terms.
  - $x+x+x+y+1+1$ is a basic term which can also be written as $3x+y+2$
- The Atomic formulas are equalities and inequalities between basic terms.
  - $x-2y = -2$ is an atomic formula with equality.
  - $x-2y \leq -2$ is an atomic formula with inequality.
- The Formulas are first-order formulas build on atomic formulas using the connectives $\wedge$(conjunction), $\vee$(disjunction), $\neg$(negation), $\exists x$ (existential quantification), $\forall x$ (universal quantification).
  - $\forall x, \exists y.(x=2y \vee x=2y+1)$ is a formula.
  - $\top$ is the empty conjunction (valid formula).
  - $\perp$ is the empty disjunction (unsatisfiable formula).

Presburger Arithmetic (Contd.)

- The Free variables of a formula φ are defined as:
  - $FV(\varphi)$ = variables which are not quantified in φ
  - $FV(\varphi_1 \vee \varphi_2) = FV(\varphi_1) \cup FV(\varphi_2)$
  - $FV(\exists x.\varphi) = FV(\varphi) \setminus \{x\}$

- The interpretation domain of the formulas is the set of natural numbers $\mathbb{N}$, with 0, 1, +, =, ≤ having their usual meaning.

- A solution of the formula φ is an assignment of $x_1, x_2, \ldots, x_n$ in $\mathbb{N}$ which satisfies φ, where $FV(\varphi)=\{x_1,x_2,\ldots,x_n\}$
  - $(x \mapsto 2, y \mapsto 1)$ is one solution of $x+3 = 4y+1$
  - $(x \mapsto 4)$ is one solution of $\exists y.(x=2y)$

Sets of Integers recognized by Finite Automaton

- Every Natural number can be written as a word in base-k (k≥1) over the alphabet set {0,1,…,(k-1)}
  - 29 in base-2 is written as 0011101

- Since we want to read the word from LSB to MSB, we write the word from right to left, with LSB in left and MSB in right.
  - 29 in base-2 is written as 1011100

Integers recognized by Finite Automaton(Contd.)

- There is a mapping from {0,1,…,(k-1)}* to **N** denoted by $\sim^k$ which maps each word in base-k to its corresponding Natural number.
  - $\widetilde{101110}^{2}$ maps to 29

- n-tuples of natural number can be represented in base-k as a single word over the alphabet set {0,1,…,(k-1)}^{n}
  - {0,1,…,(k-1)}^{3} set is {0,1,…,(k-1)} x {0,1,…,(k-1)} x {0,1,…,(k-1)}
  - (13,29,6) (base-10) can be written in base-2 over the alphabet set {0,1}^{3}

as: $$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Presburger Formula to Finite Automata

1. Build automaton for the atomic formulas in the given formula
   - Automata Associated with equalities
   - Automata Associated with inequalities

2. Use these automaton to create the automata for the given formula

Automata associated with equality

- For every atomic formula

  $a_1x_1 + a_2x_2 + \ldots + a_nx_n = b$ (where $a_1, a_2, \ldots, a_n, b \in \mathbb{Z}$)

  we construct the automaton by repeating the following inference rule until no more states or transitions can be added:

$$\frac{q_c \in Q}{q_d \in Q,\ (q_c, \theta, qd) \in \delta} \quad if \begin{cases} a_1\theta_1 + \cdots + an\theta_n \ {=_2}\ c \\ d = \dfrac{c - a_1\theta_1 - \cdots - an\theta_n}{2} \\ \theta = (\theta_1, \ldots, \theta_n) \end{cases}$$

where, $\theta_i \in \{0,1\}$

clearly, $\theta \in \{0,1\}^n$

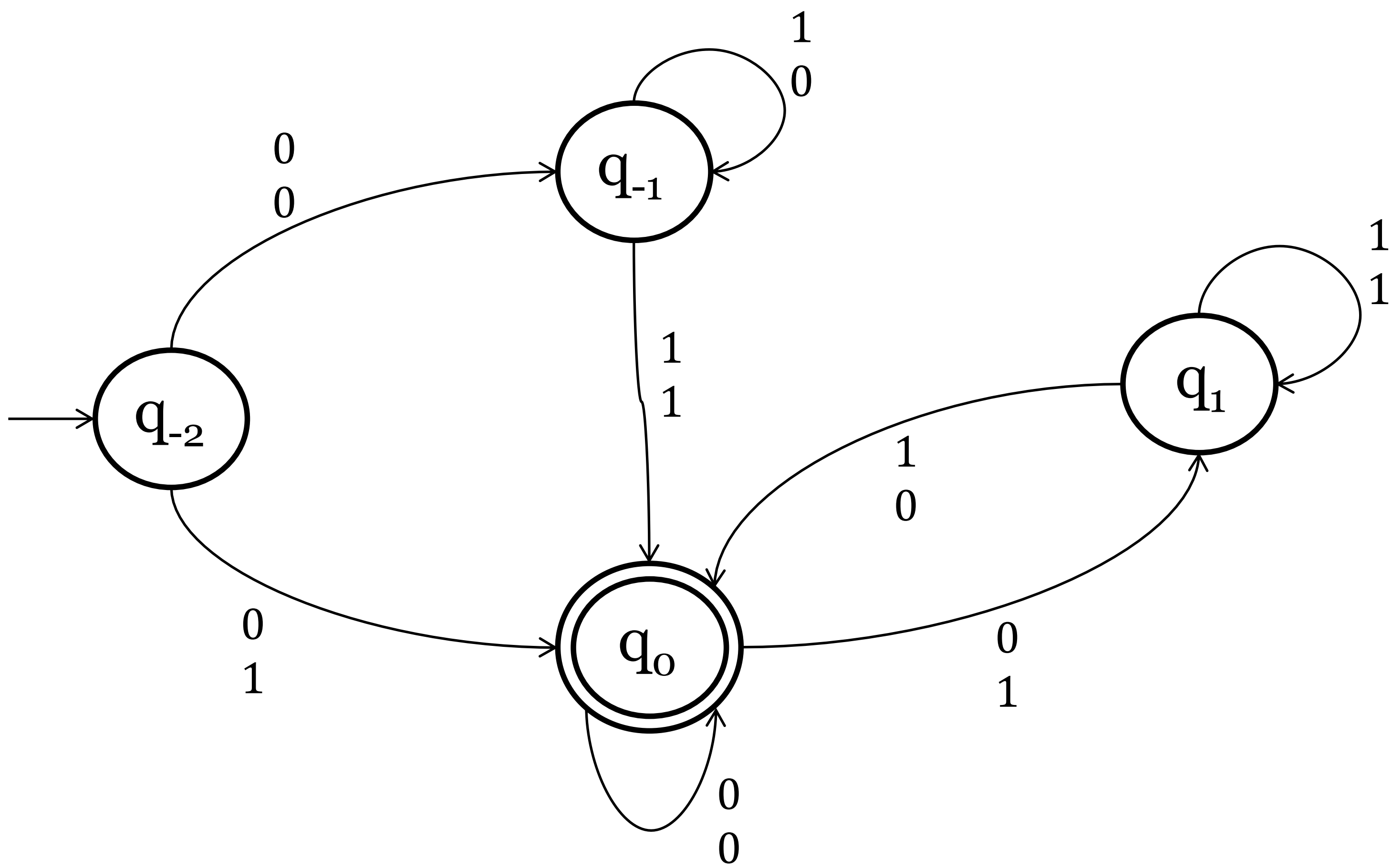
Automata associated with equality (Contd.)

- The resulting automata would be $(Q, \Sigma, \delta, q_b, F)$
  - $Q = \{q_{i's}$ computed by the inference rule$\}$
  - $\Sigma = \{0,1\}^n$
  - $\delta \subseteq (Q \times \Sigma \times Q)$ computed by the reference rule
  - q_b is the initial state
  - $F = \begin{cases} \{q_0\}, & \text{if } q_0 \in Q \\ \emptyset, & \text{otherwise} \end{cases}$

Automata associated with equality (Example)

- Consider the equation : x-2y = -2
  - We have b=-2 , so the initial state $q_{-2} \in Q$
- Let c=b
- Compute the solutions $\theta=(x \mapsto \theta_1, y \mapsto \theta_2)$ of the equation: $x\text{-}2y =_2 c$
  - The solutions are $\{(x \mapsto 0, y \mapsto 0), (x \mapsto 0, y \mapsto 1)\}$ for c=-2
- Compute the new states for each θ and the transition to them from c
  - For $\theta=(x \mapsto 0, y \mapsto 0)$, d = $\frac{c-0+0}{2} = \frac{-2-0-0}{2} = -1$
    - add $q_{-1} \in Q$
    - add $(q_{-2}, \begin{smallmatrix}0\\0\end{smallmatrix}, q_{-1}) \in \delta$
- Similarly calculate the state and transition for other θ's
- Repeat for each new state until no more new states can be added.

Automaton for: x-2y=-2



Getting solutions from the automaton

- For every successful run of the automaton, we get one solution.
- For example, for the successful run $\rho = q_{-2}\, q_{-1}\, q_{-1}\, q_0\, q_1\, q_0$

$$q_{-2} \xrightarrow{\frac{0}{0}} q_{-1} \xrightarrow{\frac{1}{0}} q_{-1} \xrightarrow{\frac{1}{1}} q_0 \xrightarrow{\frac{0}{1}} q_1 \xrightarrow{\frac{1}{0}} q_0$$

  - $x = \widetilde{01101}^{2} = 22$
  - $y = \widetilde{00110}^{2} = 12$
  - This is one solution to the equation: $x-2y=-2$

Observations on the automaton

- For an automaton for: $a_1x_2+a_2x_2+\ldots+a_nx_n=b$

1. The absolute value of the states are bounded by:

   $$max(|b|, \sum_{i=1}^{n} |ai|)$$

2. Any successful run starting from q_c gives a solution to:

   $$a_1x_2+a_2x_2+\ldots+a_nx_n=c$$

Automata associated with inequality

- Every inequality can be written in the form of:

  $a_1x_1 + a_2x_2 + \dots + a_nx_n \le b$ (where $a_1, a_2, \dots, a_n, b \in \mathbb{Z}$)

  we construct the automaton by repeating the following inference rule until no more states or transitions can be added:

$$\frac{q_c \in Q}{q_d \in Q,\ (q_c, \theta, qd) \in \delta} \quad if \begin{cases} d = \lfloor \dfrac{c - a_1\theta_1 - \cdots - an\theta_n}{2} \rfloor \\ \theta = (\theta_1, \dots, \theta_n) \end{cases}$$

  where, $\theta_i \in \{0,1\}$

  clearly, $\theta \in \{0,1\}^n$

  we apply this formula for all possible combinations of θ

Automata associated with inequality (Contd.)

- The resulting automata would be $(Q, \sum, \delta, q_b, F)$
  - Q = {$q_{i\text{'s}}$ computed by the inference rule}
  - $\sum = \{0,1\}^n$
  - $\delta \subseteq (Q \times \sum \times Q)$ computed by the reference rule
  - q_b is the initial state
  - $F = \{q_c | c \geq 0\}$

Automata associated with inequality (Example)

- Consider the equation : $x-2y \leq -2$
  - We have $b=-2$, so the initial state $q_{-2} \in Q$
- Let $c=b$
- Take $\theta=(x \mapsto \theta_1, y \mapsto \theta_2)$, where $\theta_1, \theta_2 \in \{0,1\}$
  - $\theta \in \{(x \mapsto 0, y \mapsto 0), (x \mapsto 0, y \mapsto 1), (x \mapsto 1, y \mapsto 0), (x \mapsto 1, y \mapsto 1)\}$
- Compute the new states for each θ and the transition to them from c
  - For $\theta=(x \mapsto 0, y \mapsto 0)$, $d = \lfloor \frac{c-0+0}{2} \rfloor = \lfloor \frac{-2-0-0}{2} \rfloor = -1$
    - add $q_{-1} \in Q$
    - add $(q_{-2}, \begin{smallmatrix}0\\0\end{smallmatrix}, q_{-1}) \in \delta$
- Similarly calculate the state and transition for other θ's
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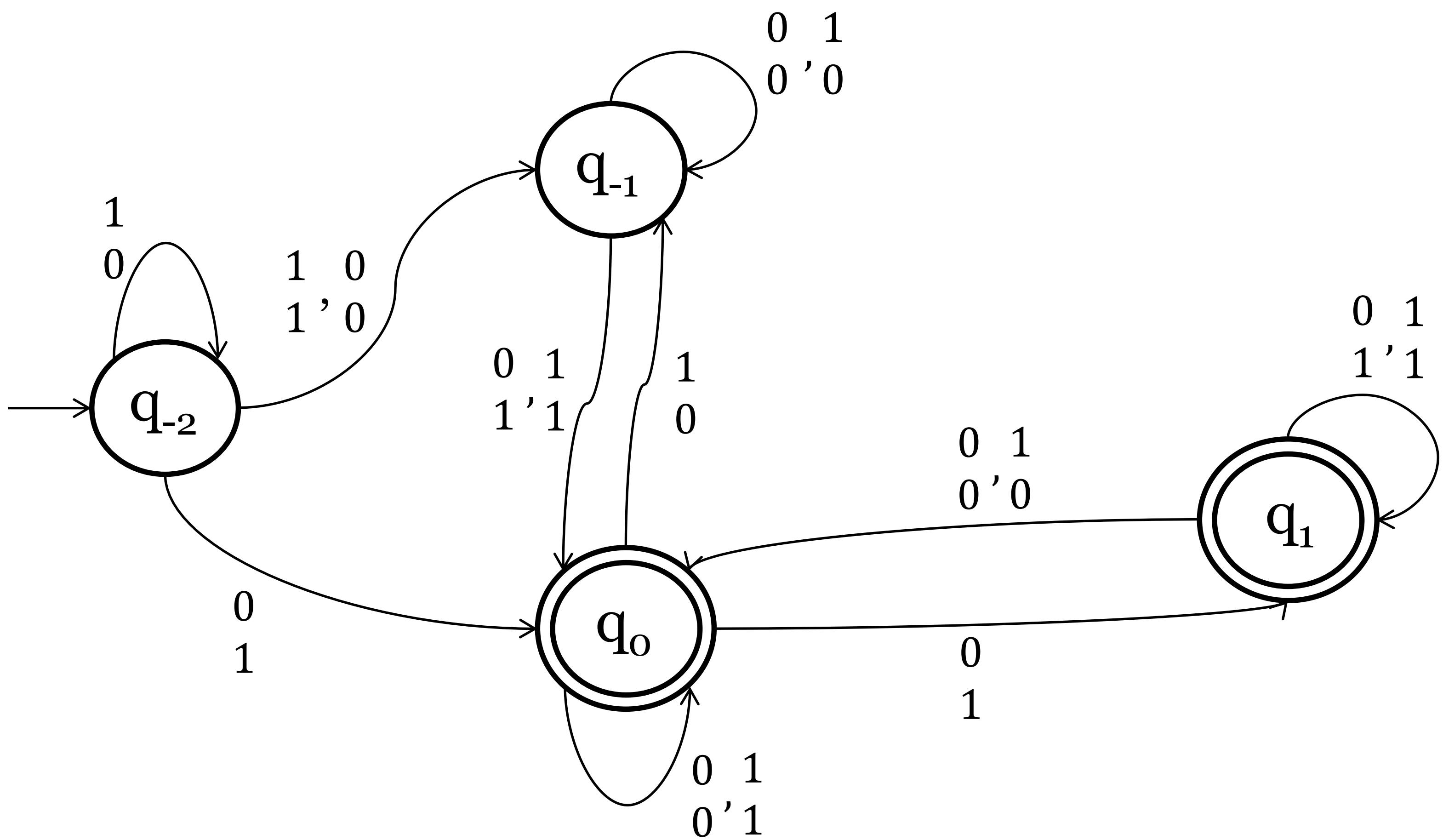
Automaton for: x-2y≤-2



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- For example, for the successful run $\rho = q_{-2}\, q_{-1}\, q_{-1}\, q_0\, q_1\, q_0$

$$q_{-2} \xrightarrow{\frac{1}{1}} q_{-1} \xrightarrow{\frac{1}{0}} q_{-1} \xrightarrow{\frac{0}{1}} q_0 \xrightarrow{\frac{0}{1}} q_1 \xrightarrow{\frac{1}{0}} q_0$$

  - $x = \widetilde{11001}^{2} = 19$
  - $y = \widetilde{10110}^{2} = 13$
  - This is one solution to the equation: $x-2y \leq -2$

Closure Properties and automata for arbitrary formula

- Let $A_\varphi(x_1,x_2,\ldots,x_n)$ be an automaton over alphabet $\{0,1\}^n$ which accepts the solutions of the formula φ.
  - Then $FV(\varphi) \subseteq \{x_1,x_2,\ldots,x_n\}$
  - $x_i \notin FV(\varphi) \Rightarrow$ accepting or not accepting a word in the automaton will not depend on the i^{th} track.

- **Exercise:**

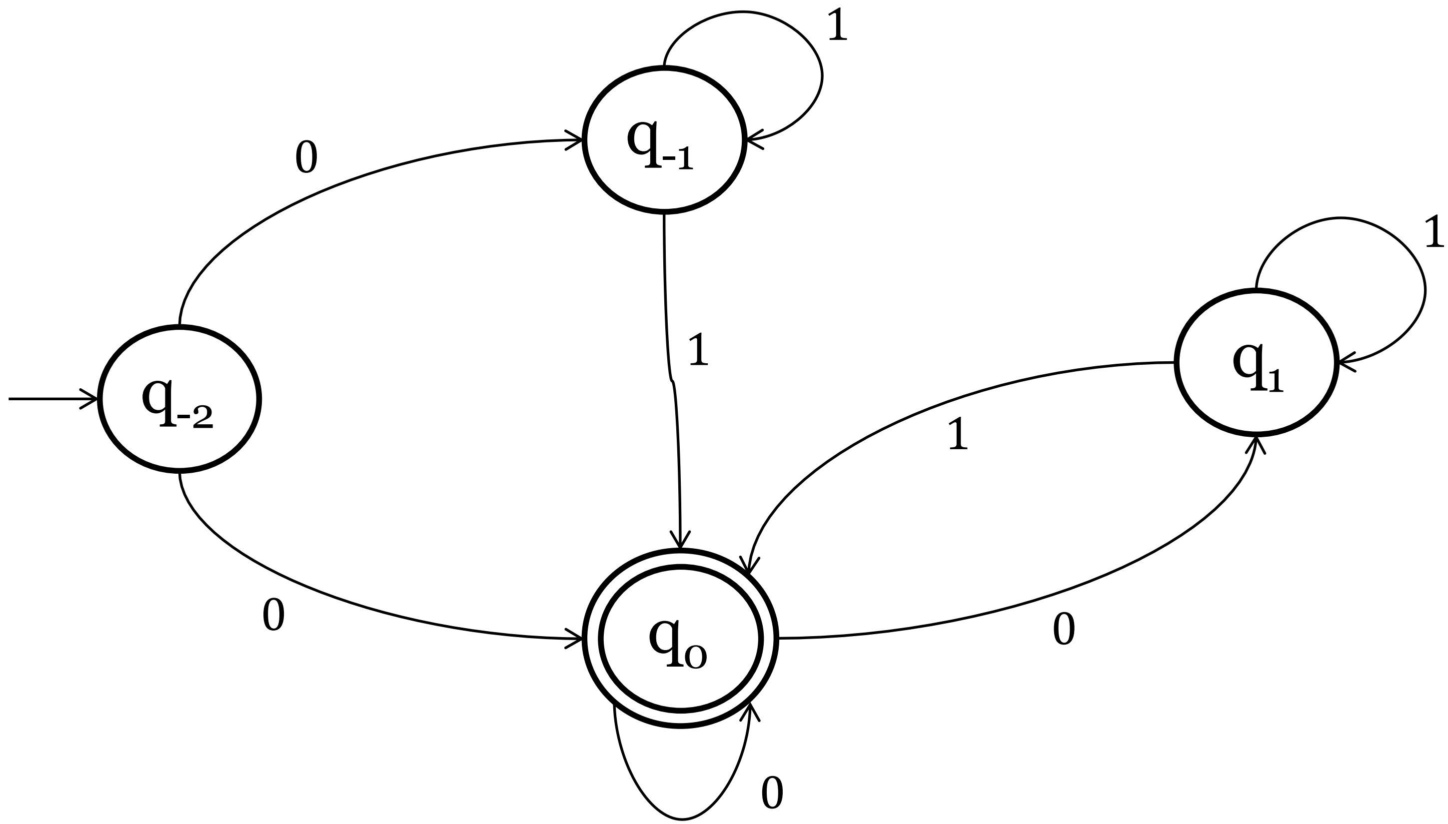
  How to compute $A_\varphi(x_1,x_2,\ldots,x_n,y)$? Given $A_\varphi(x_1,x_2,\ldots,x_n)$ and $y \notin FV(\varphi)$.

Automata for arbitrary formulas (Contd.)

- Given 2 Automata $A_{\varphi_1}(\vec{x})$ and $A_{\varphi_2}(\vec{x})$
  - $A_{\varphi_1 \wedge \varphi_2}(\vec{x})$ can be computed by the intersection of the two given automaton
  - $A_{\varphi_1 \vee \varphi_2}(\vec{x})$ can be computed by the union of the two given automaton
  - We have to make sure that $\vec{x}$ contains $FV(\varphi_1) \cup FV(\varphi_2)$, which can be done easily.
- Given Automata A_φ, we can construct $A_{\neg\varphi}$ by taking complement of A_φ.
- $A_{\exists x.\,\varphi}$ can be computed from A_φ by projection, i.e. in A_φ simply forgetting about the track corresponding to x in the transition labels.
- Since $\forall x.\,\varphi$ is logically equivalent to $\neg\exists x.\,\neg\varphi$, we can easily compute $A_{\forall x.\,\varphi}$ using A_φ

An example of Projection

- Automaton for: $\exists y. (x-2y = -2)$



Deciding satisfiability of a formula

- We can apply induction on the Presburger formula φ to compute an automaton A'_{φ} which accepts the set of solutions of φ.

- By computing A'_{φ} , deciding satisfiability of φ reduces to deciding the emptiness of the language of A'_{φ}, which can be done in linear time w.r.t number of states.

Thank you

Questions...